



AWS Cloud Practitioner

1. Module Information

Module Title	AWS Cloud Practitioner
Duration	8 Sessions
Total Hours	24 Hours

2. Module Description

This module introduces the essential AWS cloud concepts required for deploying and managing AI/ML solutions. Learners will gain knowledge of core AWS services such as IAM, EC2, S3, and CloudWatch, as well as application services including Polly, Lex, Transcribe, Rekognition, and Comprehend. The module emphasizes secure cloud practices, model deployment using containers and APIs, and AWS monitoring tools. By the end of the module, students will be able to deploy ML models securely and efficiently on AWS infrastructure.

3. Learning Outcomes

- 1 Understand AWS application services for AI and machine learning.
- 2 Gain practical knowledge of AWS SageMaker for training and deploying models.
- 3 Apply basic AWS security concepts including IAM, S3, EC2, and security groups.
- 4 Deploy ML models using Docker containers and FastAPI.
- 5 Monitor deployed models with AWS logging and monitoring tools (CloudTrail, CloudWatch).



4. Session Breakdown

Session	Topic	Hours	LOs
1	Introduction to AWS Cloud Concepts	3	1
2,3	AWS Application Services	6	1
4	AWS SageMaker for ML Development & Deployment	3	2
5	S3, EC2 & Security Groups	3	3
6,7	Deploying ML Models with Docker Containers	6	4
8	Monitoring & Logging with CloudTrail and CloudWatch	3	5

Sessions Highlights:

- Hands-on AWS labs for key services (Lex chatbot, Polly text-to-speech, Rekognition image analysis).
- Building and deploying ML pipelines in SageMaker.
- Practical implementation of IAM policies for secure ML projects.
- End-to-end deployment of a model using Docker + FastAPI.
- Monitoring workflows with CloudTrail event logs and CloudWatch dashboards.

7. Teaching Methods

#	Method
1	Guided hands-on labs in AWS console and CLI.
2	Team-based deployment exercises.
3	Case studies of cloud-based ML applications.



8. Assessment Methods

#	Method	Marks
1	Quizzes on AWS concepts	30%
2	Hands-on AWS Labs	30%
3	Deployment Project	40%